

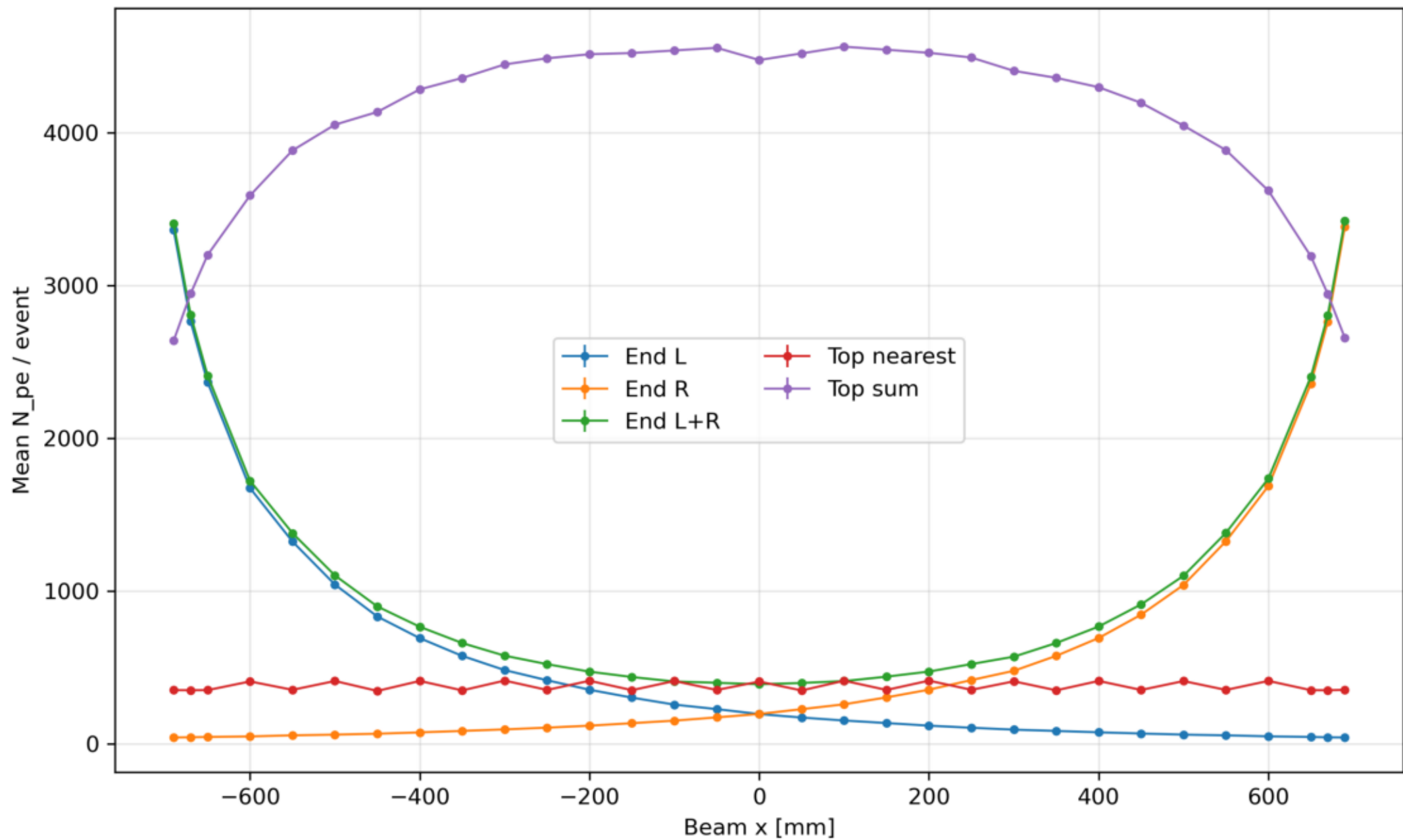
EXEC_08 - EXEC_07 EndTop photon budget

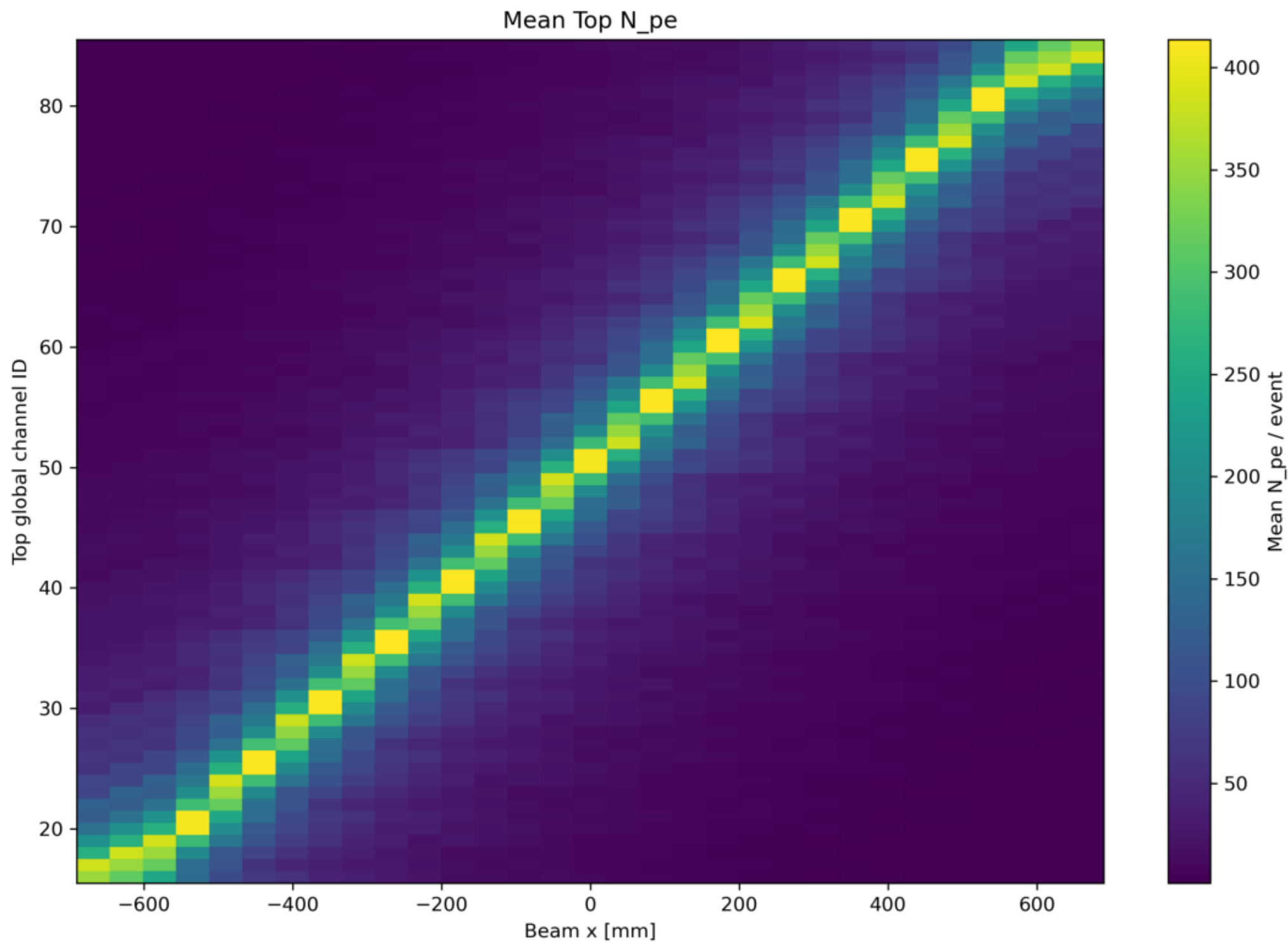
31 positions x 2000 events, 86 channels, EJ-204 OPSC-101, Broadcom AFBR-S4N66P024M

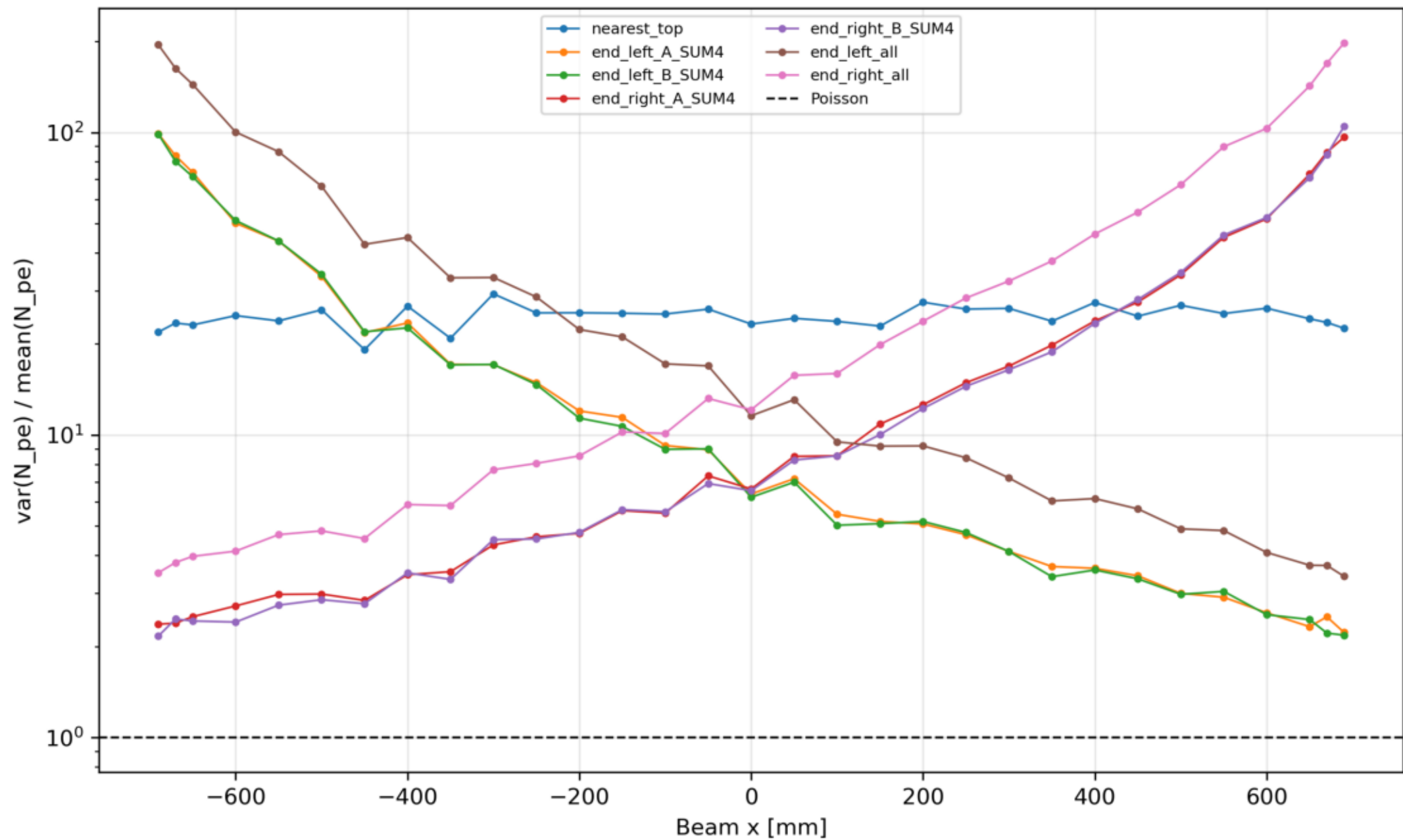
Intrinsic simulation: /sipm/jitterSigma 0

- Effective attenuation: left 33.14 +/- 0.03 cm; right 33.06 +/- 0.03 cm. These include light extraction through 70 Top windows and are not expected to equal
- Effective propagation velocity from mean all-photon time: left 27.68 +/- 0.27 cm/ns; right 27.67 +/- 0.27 cm/ns.
- Typical nearest-Top var/mean: median 24.81, range 19.14-29.27. N_{pe} is not forced to Poisson.
- Intrinsic End SUM4 sigma_group= sigma(DeltaT)/sqrt(2): mean 148.28 ps, range 12.10-437.32 ps. No SPTR/electronics jitter.
- sigma(t_avg) mean 142.49 ps, range 93.20-225.36 ps.

Top uses 70 simulated SiPM elements and does not replicate the 32-SiPM hardware setup.

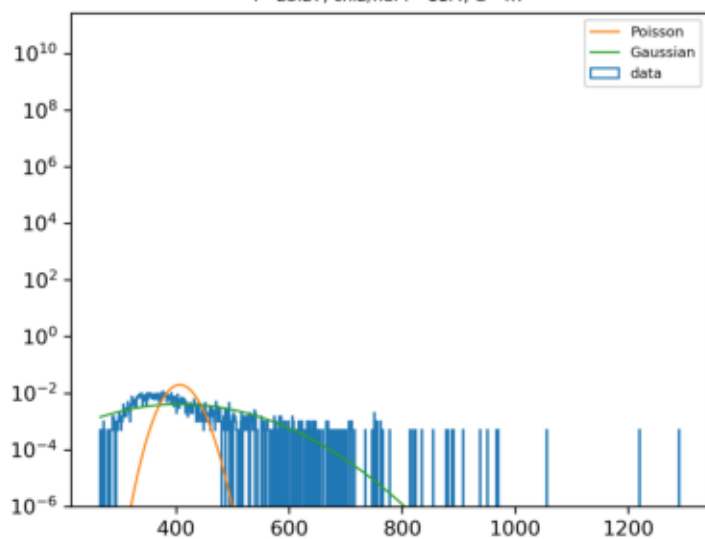




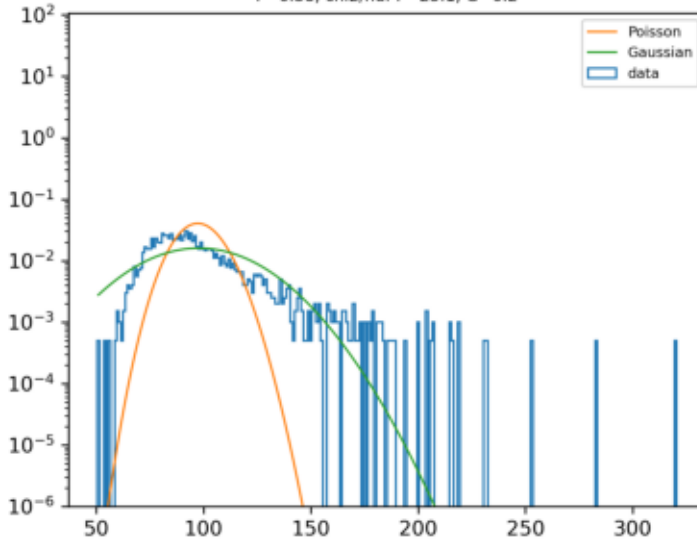


Poisson tail check, x=0 mm

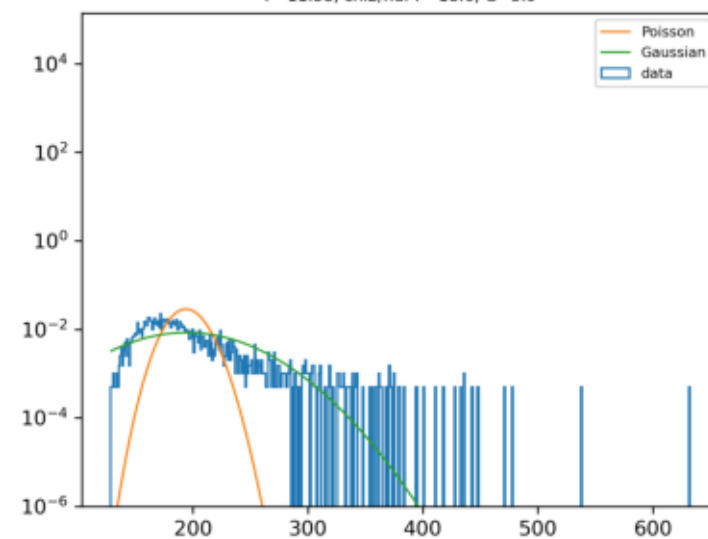
nearest_top
F=23.27, χ^2/ndf P=11.4, G=4.7

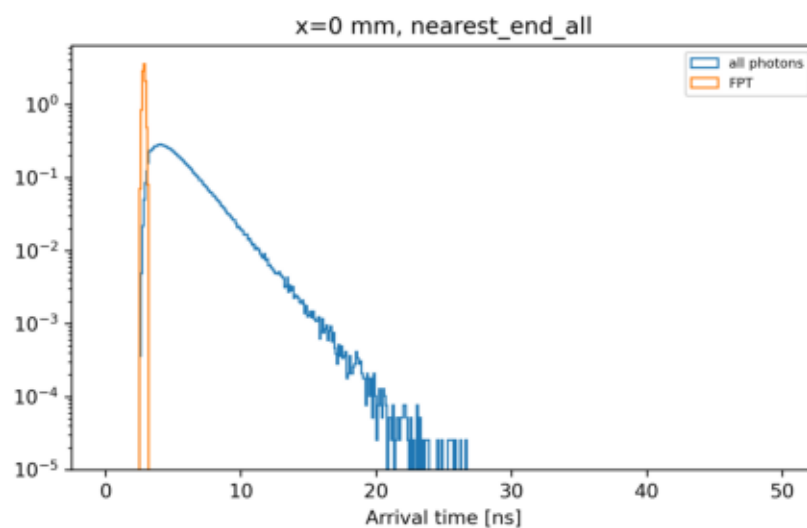
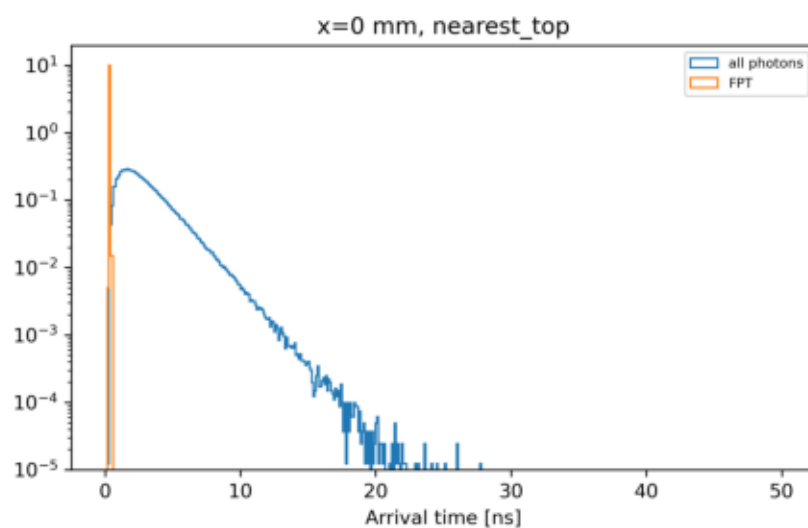
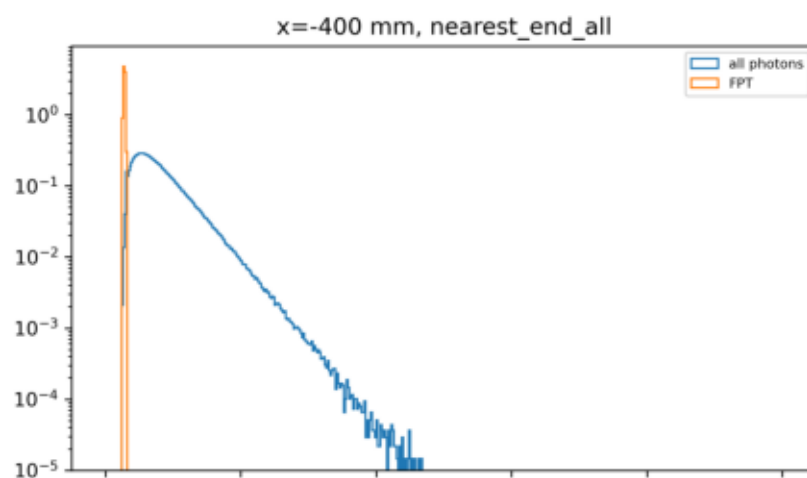
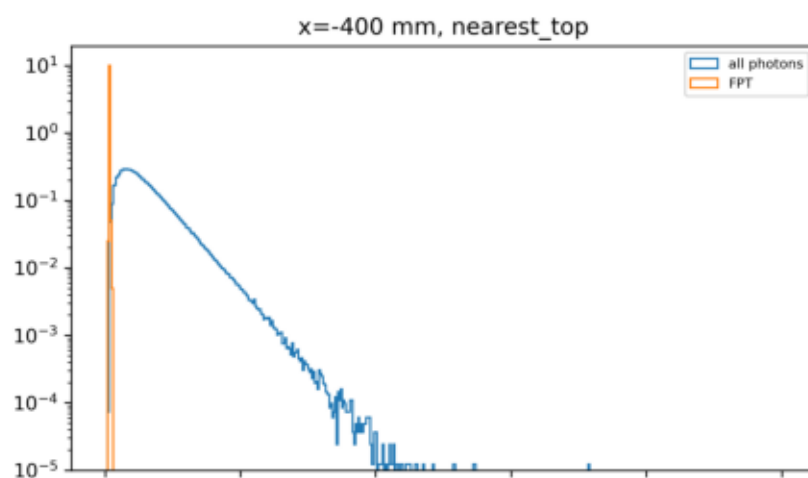
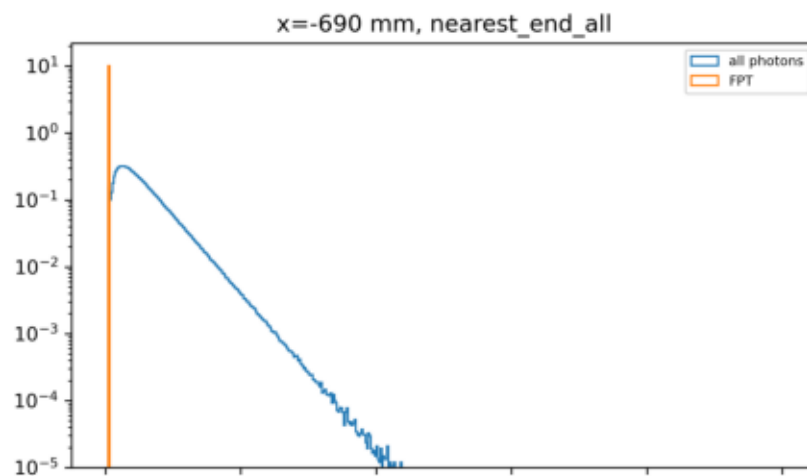
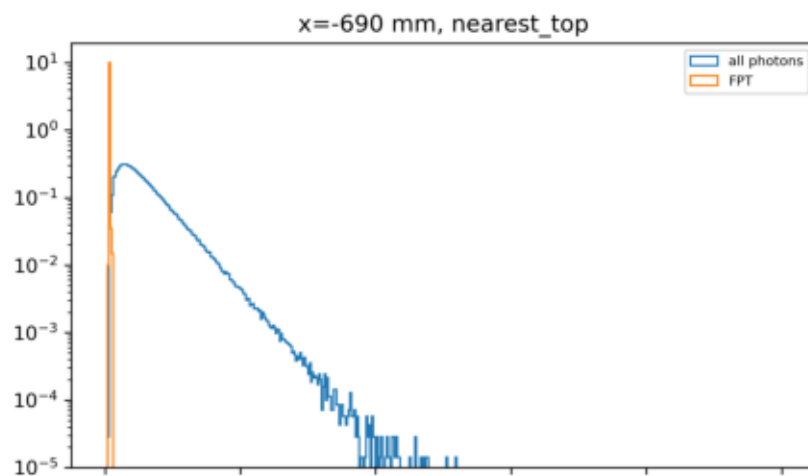


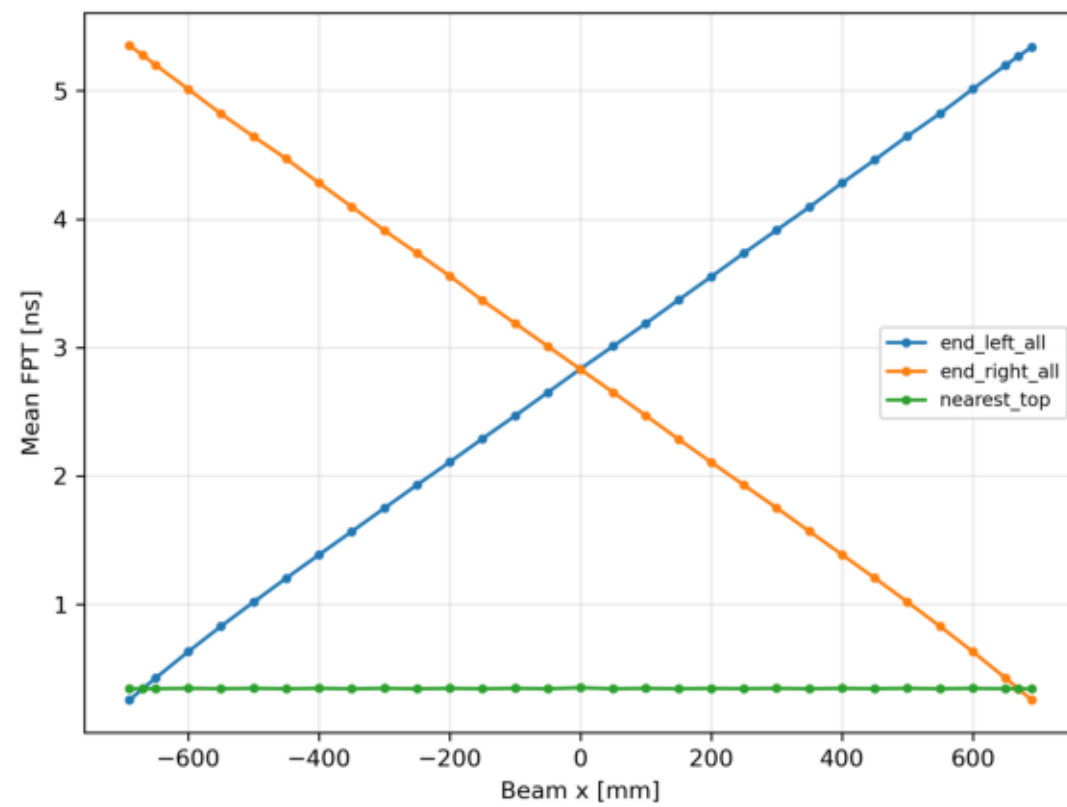
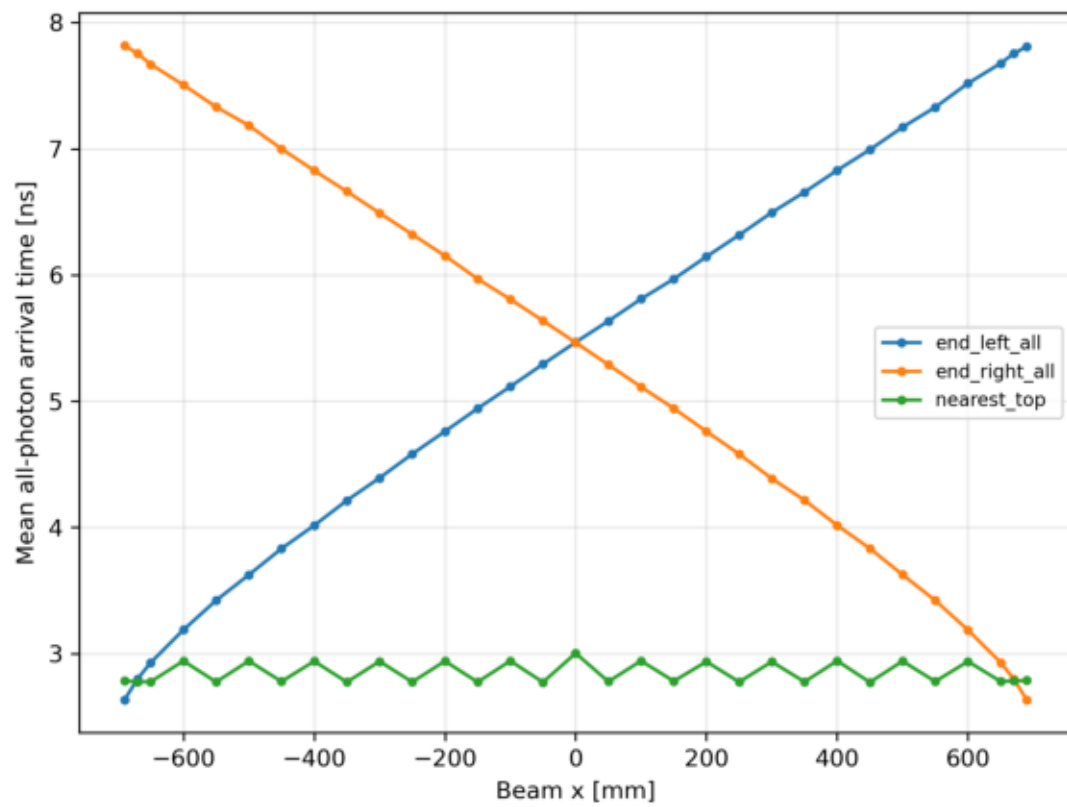
nearest_end_SUM4
F=6.38, χ^2/ndf P=25.1, G=6.2



nearest_end_all
F=11.58, χ^2/ndf P=18.6, G=5.9







Intrinsic End timing, no SPTR/electronics jitter

